2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSML-0902

(Probability and Statistics)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2**, and ***any four*** questions from the rest.

1. Answer ***any five*** questions. 2×5

(a) What do you mean by Categorical Data? Give one example.

(b) What is the probability of throwing 16 with three dice?

(c) Given $p(x)=\begin{cases}\frac{x}{21}; x=1,2,3,4,5,6\\ 0 \quad \text{otherwise}\end{cases}$

prove that p(X) is a Probability Mass Function.

(d) How can we calculate the Conditional probability?

(e) State the Central Limit Theorem.

(f) What do you mean by Estimation? Give one example where point estimation is not suitable.

(g) What is Confidence Interval? Explain it over the Normal Curve.

2. Answer ***any five*** questions : 4×5

(a) Find the Probability of drawing five cards among which three of a kind is chosen from a deck of well-shuffled cards (52 cards).

(b) For a specific married couple the probability that the husband watches the show is 80%, the probability that his wife watches the show is 65%, while the probability that they both watch the show is 60%. If the husband is watching the show, what is the probability that his wife is also watching the show?

(c) Find the mean and variance of the distribution : $f(x)=8x\,(1-x)$; where $0 \leq x \leq 1$.

(d) A certain calculating machine uses only the digits 0 and 1. It is supposed to transmit one of these digits through several stages. However, at every stage, there is a probability p that the digit that enters this stage will be changed when it leaves and a probability $q=1-p$ that it won't. Form a Markov chain to represent the process of transmission by taking as stated the digits 0 and 1. What is the matrix of transition probabilities?

(e) The Joint probability distribution of two random variables X and Y is given by :

$$f(x,y)=\begin{cases}\frac{1}{8}(6-x-y) & 0<x<2, 2<y<4\\ 0 & \text{otherwise}\end{cases}$$

Find $P(X+Y<3)$.

(f) Derive the mean for Normal Distribution.

(g) I want to rent an unfurnished apartment in Bhubaneswar. The mean monthly rent for a random sample of 60 apartments advertised on HOUSING.COM (a website that lists apartments for rent) is 10000. Assume a population standard deviation of 2000. Construct a 95% confidence interval. (Z=1.96)

3. (a) Two random variables X and Y have the following joint probability density function :

$$f(x,y)=\begin{cases}2-x-y; & 0\le x\le 1, 0\le y\le 1\\ 0 & \text{otherwise}\end{cases}$$

Find

(i) Conditional density function

(ii) Var (X) and Var (Y).

(b) The chance that System A will diagnose error in software module correctly is 60%. The chance of a crash even after the correct error diagnosis is 40% and the chance of crash without diagnosis is 70%. System A crashed due to some error in module. What is the probability that the error was diagnosed correctly? (3+2)+5

4. (a) I have four umbrellas, some at home, some in the office. I keep moving between home and office. I take an umbrella with me only if it rains. If it does not rain I leave the umbrella behind (at home or in the office). It may happen that all umbrellas are in one place, I am at the other, and it starts raining and must leave, so I get wet. If the probability of rain is p,

(i) Draw the State Transition Diagram.

(ii) What is the probability that I get wet?

(iii) Current estimates show that p = 0.6 in Kolkata.

(b) How many umbrellas should I have so that, if I follow the strategy above and p = 0.6 the probability I get wet is less than 0.1? 6+(2+2)

5. (a) A machine has two parts and it bases the decision on calculations done independently by each of parts. The probability that the machine makes a mistake in its processing is 0.005. The probability that a part gives wrong calculation is 0.3. Assuming that the mistakes made by the machine itself are independent of the calculations given by the parts, find the probability that the machine takes a wrong decision.

(b) The population standard deviation for the age of CU students is 18 years. If we want to be 95% confident that the sample mean age is within 2 years of the true population mean age of CU students, how many randomly selected CU students must be surveyed? 5+5

6. (a) A roulette wheel has 38 slots. Thirty-six slots are numbered from 1 to 36; half of them are red and half are black. The remaining two slots are numbered 0 and 00 and are green. In an INR100 bet on red, the bettor pays INR 100 to play. If the ball lands in a red slot, he receives back the money he bet plus additional money (100INR). If the ball does not land on red he loses his money. Let X denote the net gain to the bettor on one play of the game.

(i) Construct the probability distribution of X.

(ii) Compute the mean and variance of X.

(b) There is a 0.9986 probability that a randomly selected 30 years old WB citizen will live throughout the year. An insurance company charges INR 161 for insuring the male will live through the year, with a 100,000 payout if he dies during this time. What is the expected value of this policy to the insurance company? (2+4)+4

7. (a) The continuous random variable X has the following probability density function :

$$f(x) = \begin{cases} \frac{2}{29}(x+1) & \text{for } 1 < x < 4 \\ \frac{2}{29} & \text{for } 4 < x < 8 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function.

(ii) Find $P(2 < X < 5)$.

(iii) Find the value of a for which $P(X > a) = 0.1$.

(b) In a binomial distribution (n, p) calculate the variance. (3+2+3)+2

8. (a) Let $X_1, X_2, \ldots X_n$ be a random sample from a population with pdf

$$f(x \mid \theta) = (\theta + 1)\, x^{\theta},\ 0 < x < 1.$$

Find the maximum likelihood estimator and method of moments estimator of θ, when $x_1 = 0.10$, $x_2 = 0.22$, $x_3 = 0.54$, $x_4 = 0.36$.

(b) A survey found how many hours of sunshine vs. how many sunscreen creams were sold at the shop from Monday to Friday.

Hours of Sunshine at Kolkata :	2	3	5	7	9
Sunscreen cream sold :	4	5	7	10	15

Construct the Least Square regression equation from the above data. 5+5